Thorax Anatomy & Embryology: Pericardium, Heart, & CVS Development

(Overview of Pericardium and Heart
Describe the macroscopic structures of heart and pericardium

Development of CVS
List parts of primitives of heart tube & their derivatives)

1. The pericardium is a fibroserous sac enclosing the heart and the roots of the great vessels. Macroscopically, the outermost, tough, protective connective tissue layer is the:
a. Visceral pericardium
b. Epicardium
c. Endocardium
d. Myocardium
e. Fibrous pericardium
2. The inner serous pericardium is a closed sac divided into two layers. The layer that is tightly adherent to the external muscular surface of the heart is the visceral layer, also known as the:
a. Endocardium
b. Myocardium
c. Fibrous sheet
d. Epicardium
e. Mediastinal pleura
3. Bounded between the parietal and visceral layers of the serous pericardium is the pericardial cavity. In a healthy subject, this potential space contains a thin film of fluid that functions to:
a. Synthesize clotting factors
b. Lubricate the heart surfaces to minimize friction during contraction
c. Generate high-pressure arterial loops
d. Pump deoxygenated blood to the lungs
e. House the coronary sinus valves
4. Rapid accumulation of excess fluid or blood inside the rigid fibrous pericardium (such as from a stab wound) compresses the heart chambers and prevents normal filling, a life-threatening clinical condition known as:
a. Pleural effusion
b. Pneumothorax
c. Cardiac tamponade
d. Goiter collapse
e. Zenker's diverticulum
5. The serous pericardium reflects around the great vessels to form two highly unique, high-yield clinical recesses. The transverse pericardial sinus is situated horizontally posterior to the:
a. Left atrium and pulmonary veins
b. Superior and inferior venae cavae
c. Ascending aorta and pulmonary trunk (anteriorly) and anterior to the superior vena cava

d. Descending thoracic aorta
e. Left ventricle apex

6. The oblique pericardial sinus is a blind, J-shaped mucosal pocket located behind the heart, bounded laterally by the:

a. Right and left ventricles
b. Pulmonary veins and inferior vena cava
c. Ascending aorta and brachiocephalic trunk
d. Internal thoracic vessels
e. Thalamic pulvinar tract

7. High-pressure arterial blood supply to the fibrous and parietal pericardium is provided primarily by branches derived from the:

a. Internal thoracic artery (via musculophrenic and pericardiophrenic arteries)
b. External carotid artery
c. Superior thyroid artery
d. Lingual artery
e. Coronary arteries exclusively

8. The somatic sensory nerve fibers carrying pain awareness from the fibrous and parietal pericardium to the central nervous system travel within the:

a. Vagus nerves (CN X)
b. Sympathetic trunk chain
c. Recurrent laryngeal nerves
d. Phrenic nerves (explaining referred pain to the C3–C5 dermatomes over the shoulder)
e. Intercostal nerves

9. The heart is a hollow muscular pump positioned obliquely inside the chest. Macroscopically, which mediastinal compartment houses the heart and its pericardial sac?

a. Superior mediastinum
b. Anterior mediastinum
c. Middle mediastinum
d. Posterior mediastinum
e. Prevertebral space

10. The heart features a distinct apex that points downwards, forwards, and to the left. In a normal subject, the apex is formed entirely by the:

a. Right atrium
b. Right ventricle
c. Left atrium
d. Left ventricle
e. Coronary sinus

11. Clinically, the apex beat of the heart can be easily palpated on the chest wall in which precise anatomical location?

a. Left 2nd intercostal space near the sternum
b. Right 4th intercostal space
c. Left 5th intercostal space, roughly 9 cm lateral to the sternal midline (midclavicular line)
d. Xiphoid process tip
e. Sternal notch line

12. The base of the heart faces posteriorly toward the thoracic vertebrae and is formed predominantly by which chamber?

a. Left atrium (along with a small part of the right atrium)

b. Left ventricle
c. Right ventricle
d. Right atrium exclusively
e. Conus arteriosus

13. The anterior (sternocostal) surface of the heart is positioned deep to the sternum and ribs. It is formed macroscopically mainly by the:

a. Left atrium
b. Right ventricle (along with part of the right atrium and left ventricle)
c. Left ventricle apex purely
d. Inferior vena cava mouth
e. Pyramidal lobe rest

14. The interior of the right atrium consists of a smooth posterior wall and a rough anterior wall. The muscular ridges that form the rough internal appearance of the anterior wall are the:

a. Trabeculae carneae
b. Chordae tendineae
c. Pectinate muscles (musculi pectinati)
d. Papillary muscles
e. Valvulae sinus

15. Separating the smooth and rough internal segments of the right atrium is a distinct vertical muscular ridge visible on the endocardial surface, known as the:

a. Interatrial septum
b. Crista terminalis
c. Fossa ovalis
d. Conus arteriosus
e. Septomarginal trabecula

16. The smooth posterior wall of the right atrium features openings for the superior vena cava, inferior vena cava, and a major venous channel draining the heart walls, called the:

a. Pulmonary vein
b. Azygos vein
c. Coronary sinus
d. Internal jugular vein
e. Subclavian vein

17. On the smooth interatrial septum of the right atrium is a shallow, oval depression that represents an embryonic remnant of the fetal interatrial shunt. This depression is the:

a. Foramen ovale
b. Crista terminalis
c. Fossa ovalis
d. Ostium secundum
e. Sella turcica

18. The right ventricle pumps deoxygenated blood to the lungs. Its thick muscular internal walls display irregular muscular bundles and ridges classified as:

a. Pectinate muscles
b. Trabeculae carneae
c. Chordae tendineae
d. Maculae adherens

e. Zymogen plates

19. Projecting from the walls of the right ventricle are specialized conical muscles that anchor delicate, thread-like fibrous cords to the cusps of the tricuspid valve. These structures are the:

a. Pectinate muscles and cristae
b. Papillary muscles and chordae tendineae
c. Purkinje fibers
d. Trabeculae carneae sheets
e. Valvulae of Eustachius

20. A unique, isolated bundle of specialized trabeculae carneae tracks transversely across the right ventricular cavity from the interventricular septum to the base of the anterior papillary muscle. This bundle is the:

a. Crista supratentorialis
b. Chordae tendineae loop
c. Moderator band (septomarginal trabecula, carrying the right branch of the AV bundle)
d. Atrioventricular node
e. Tendon of Todaro

21. The smooth, funnel-shaped outflow tract of the right ventricle that leads directly upward into the pulmonary trunk is the:

a. Conus arteriosus (infundibulum)
b. Vestibule of aorta
c. Fossa ovalis
d. Coronary sinus valve
e. Apex cordis

22. Blood flows from the right atrium into the right ventricle through an orifice guarded by a valve featuring three specialized cusps, designated as the:

a. Bicuspid (mitral) valve
b. Aortic semilunar valve
c. Tricuspid valve
d. Pulmonary semilunar valve
e. Eustachian valve

23. The left atrium is a smooth-walled chamber that receives highly oxygenated blood from the lungs via how many total pulmonary veins?

a. Two pulmonary veins
b. Three pulmonary veins
c. Four pulmonary veins (two right, two left)
d. One single vein
e. Six pulmonary veins

24. The walls of the left ventricle are remarkably thick—approximately three times thicker than those of the right ventricle. This robust muscular architecture is essential to:

a. Receive low-pressure blood from the venae cavae
b. Pump blood against high systemic vascular resistance to the entire body
c. House the fossa ovalis membrane
d. Contain the entire coronary sinus plexus
e. Protect the pulmonary trunk arch

25. Blood passes from the left atrium into the left ventricle through an orifice guarded by a high-pressure valve featuring two distinct cusps, known as the:

a. Tricuspid valve
b. Bicuspid (mitral) valve
c. Aortic valve
d. Pulmonary valve
e. Thebesian valve

26. The exit portals of the right and left ventricles lead into the pulmonary trunk and ascending aorta, respectively. These high-yield outflow tracts are guarded by the:

a. Atrioventricular valves
b. Tricuspid and mitral sheets
c. Semilunar valves (pulmonary and aortic valves, each featuring three pocket-like cusps)
d. Eustachian and Thebesian flaps
e. Mitral valve apparatus exclusively

27. The arterial blood supply to the heart muscle itself is delivered via the right and left coronary arteries. These primary vessels originate directly from the:

a. Arch of the aorta
b. Brachiocephalic trunk
c. Right and left aortic sinuses (of Valsalva) located immediately superior to the aortic valve cusps
d. Internal thoracic artery
e. Subclavian artery first part

28. The left coronary artery branches quickly after its origin. Its high-yield branch that runs down the anterior midline of the heart toward the apex is the:

a. Marginal artery
b. Anterior interventricular artery (Left Anterior Descending / LAD)
c. Circumflex artery
d. Posterior interventricular artery
e. Nodal branch

29. Occlusion of a coronary artery (usually due to atherosclerosis) cuts off blood flow and causes localized coagulative necrosis of the myocardium, a highly dangerous condition termed:

a. Angina pectoris transiently
b. Pericarditis
c. Myocardial infarction (heart attack)
d. Cardiac tamponade shift
e. Coronary sinus thrombosis

30. The cardiac conduction system regulates the heartbeat. The primary pacemaker of the heart, which initiates each electrical wave of contraction under normal conditions, is the:

a. Atrioventricular (AV) node
b. Bundle of His
c. Purkinje fiber network
d. Sinoatrial (SA) node (located at the upper end of the crista terminalis in the right atrium)
e. Interventricular septal block

31. Let's shift completely to the embryological development of the CVS. The cardiovascular system begins its formation in the middle of the third week. The heart develops from a specialized layer of mesenchyme known as the:

a. Paraxial mesoderm
b. Splanchnic (visceral) lateral plate mesoderm

c. Intermediate mesoderm
d. Somatic mesoderm
e. Surface ectoderm sheets

32. Angiogenic clusters fuse to form a pair of longitudinal endothelial channels that later merge across the midline during embryonic folding to form a single, vital:

a. Primary pharyngeal pouch
b. Laryngotracheal groove
c. Primitive heart tube
d. Aortic arch system
e. Interatrial septum

33. As the primitive heart tube develops, it displays a series of distinct expansions and constrictions along its length. The cranial-most segment that acts as the outflow tract connection to the aortic arches is the:

a. Sinus venosus
b. Bulbus cordis
c. Truncus arteriosus
d. Primitive ventricle
e. Primitive atrium

34. Situated immediately caudal to the truncus arteriosus is a segment that forms the smooth outflow parts of both adult ventricles, known as the:

a. Sinus venosus
b. Conus cordis and bulbus cordis
c. Atrium segment
d. Interatrial septum element
e. Vitelline plexus

35. The single primitive ventricle segment of the heart tube undergoes extensive expansion and remodeling to give rise to the bulk of the adult:

a. Right and left atria
b. Right and left ventricles (primarily the muscular parts)
c. Superior and inferior venae cavae
d. Ascending aorta and pulmonary trunk
e. Coronary sinus matrix

36. The single primitive atrium segment expands and loops ventrally to form the rough, trabeculated portions of both the adult:

a. Ventricles
b. Pulmonary veins
c. Right and left atria (including the auricles)
d. Aortic sinuses
e. Semilunar valves

37. The caudal-most chamber of the primitive heart tube receives all early embryonic venous channels (vitelline, umbilical, and cardinal veins). This chamber is the:

a. Truncus arteriosus
b. Primitive ventricle
c. Bulbus cordis
d. Sinus venosus
e. Conus cordis

38. The sinus venosus consists of a central body and right and left horns. During development, the right horn expands significantly and becomes completely incorporated into the wall of the right atrium to form its:
a. Rough anterior wall with pectinate muscles
b. Smooth posterior wall (sinus venarum)
c. Interatrial septum
d. Fossa ovalis rim
e. Tricuspid valve cusps
39. The left horn of the sinus venosus undergoes a different developmental pathway. It shifts across the back of the heart and regresses to form the adult:
a. Superior vena cava
b. Inferior vena cava
c. Coronary sinus (and the oblique vein of the left atrium)
d. Arch of the aorta
e. Left pulmonary artery
40. Because the heart tube grows rapidly within the confined pericardial cavity, it undergoes a crucial developmental movement, twisting downwards and to the right to form an S-shaped configuration. This process is:
a. Lateral transposition
b. Embryonic invagination
c. Cardiac looping (d-looping)
d. Septation translation
e. Rotation of the foregut
41. Partitioning of the atrioventricular canal into right and left openings during the fourth and fifth weeks is driven by the proliferation and fusion of a pair of mesenchymal masses called the:
a. Septum primum sheets
b. Atrioventricular (AV) endocardial cushions
c. Tracheoesophageal ridges
d. Bulbar ridges
e. Spiral septa
42. The division of the primitive atrium begins with the downward growth of a thin, crescent-shaped membrane from the roof of the chamber toward the endocardial cushions. This membrane is the:
a. Septum secundum
b. Septum primum
c. Septum spurium
d. Aorticopulmonary septum
e. Interventricular septum
43. Before the septum primum fuses completely with the endocardial cushions, a series of small perforations form in its upper segment, coalescing to create a vital opening called the:
a. Ostium primum
b. Ostium secundum (foramen secundum)
c. Foramen ovale
d. Interventricular foramen
e. Foramen cecum

44. A second, thicker muscular membrane grows downwards from the atrial roof immediately to the right of the septum primum, but it stops short of the cushions, leaving a persistent oval aperture named the:
a. Ostium secundum
b. Foramen ovale
c. Ductus arteriosus
d. Foramen magnum
e. Interatrial cleft
45. In the fetal circulation, the foramen ovale allows highly oxygenated placental blood entering the right atrium via the inferior vena cava to pass directly into the:
a. Right ventricle
b. Pulmonary trunk to reach the lungs
c. Left atrium (shunting blood away from the uninflated lungs)
d. Superior vena cava
e. Coronary sinus lumen
46. Division of the truncus arteriosus and bulbus cordis into the separate ascending aorta and pulmonary trunk is achieved by the formation of a spiral, twisted partition called the:
a. Septum primum
b. Aorticopulmonary (spiral) septum
c. Interventricular septum
d. Endocardial cushion bridge
e. Tracheoesophageal septum
47. A developmental failure or lack of spiral twisting during the formation of the aorticopulmonary septum results in a severe congenital birth defect where the aorta arises from the right ventricle and the pulmonary trunk arises from the left ventricle. This condition is:
a. Tetralogy of Fallot
b. Persistent truncus arteriosus
c. Transposition of the Great Vessels (Arteries)
d. Atrial septal defect
e. Patent ductus arteriosus
48. Tetralogy of Fallot is a classic, high-yield congenital heart syndrome caused by an unequal division of the truncus arteriosus due to anterior displacement of the aorticopulmonary septum. This syndrome consists of a tetrad of features:
a. Pulmonary stenosis, Ventricular Septal Defect (VSD), overriding aorta, and right ventricular hypertrophy
b. Atrial septal defect, mitral stenosis, patent ductus arteriosus, and left ventricular atrophy
c. Coarctation of the aorta, bicuspid aortic valve, dysphagia, and Horner's syndrome
d. Dextrocardia, situs inversus, sinusitis, and bronchiectasis
e. Complete heart block, goiter, macroglossia, and blindness
49. An embryonic shunt connects the left pulmonary artery directly to the arch of the aorta during fetal life to bypass the high-pressure lungs. This shunt is the ductus arteriosus, which normally closes at birth to become the fibrous:
a. Ligamentum venosum
b. Ligamentum arteriosum
c. Ligamentum teres
d. Median umbilical ligament

e. Fossa ovalis rim

50. If the ductus arteriosus fails to close postnatally, it remains open as a patent ductus arteriosus (PDA), a condition characterized clinically by a loud, continuous heart murmur described as a:

a. Crescendo-decrescendo systolic murmur
b. Low-pitched diastolic rumble
c. Machinery-like murmur heard throughout both systole and diastole
d. Brief presystolic click
e. Silent friction rub

Answer Key Table (Pericardium, Heart, & CVS Development)

Q 1-8	Q 9-16	Q 17-24	Q 25-32	Q 33-40	Q 41-48	Q 49-50
e	c	c	b	b	c	b
d	d	b	c	b	c	c
b	c	c	b	b	c	
c	b	b	c	b	b	
c	b	b	b	c	c	
b	c	c	b	c	b	
a	c	c	c	c	c	
d	b	b	b	a	a	